ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 4TH YEAR

FINAL YEAR PROJECT

PROJECT GUIDE · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central University · 4th year Final Year Project project guide.

Written by Academic Central for practice, following final-year design project modules at South African universities (e.g. University of Pretoria EPR 402 / EPR 405 Project) and the ECSA graduate attributes for the BEng degree. It is not a university memorandum.

How to mark: This is a typical marking rubric for a final-year design project, out of 100. The marks for each part are given on the right, with what earns them. Universities often also require a pass in each graduate attribute (GA) assessed, so a weak area cannot be carried by a strong one. Use your own university's rubric when you have it.

A. PROPOSAL, REQUIREMENTS AND SPECIFICATIONS (GA1)

A Clear problem statement and scope; requirements turned into measurable specifications, each with a test method; realistic plan, budget and risks. (10)

Excellent (8 to 10)	Good (6 to 7)	Adequate (5)	Poor (0 to 4)
Every specification is measurable and tested; scope and risks are well thought out.	Most specifications are measurable; a sound plan.	Some vague specifications; a basic plan.	Vague or untestable specifications; no plan.

B. LITERATURE STUDY AND BACKGROUND (GA2, GA9)

B Relevant existing solutions and theory, critically compared and correctly referenced; shows independent learning. (10)

Excellent (8 to 10)	Good (6 to 7)	Adequate (5)	Poor (0 to 4)
Critical comparison of good sources, clearly linked to the design.	Relevant sources, mostly described.	A few sources, little link to the design.	Few or unreliable sources; copied text.

C. ENGINEERING DESIGN (GA3)

C Concepts generated and selected with justification; detailed design from first principles (calculations, models, simulations) for the mechanical, electronic and software parts; safety, cost and impact considered. (25)

Excellent (20 to 25)	Good (15 to 19)	Adequate (13 to 14)	Poor (0 to 12)
Every major choice is justified by correct calculations or simulations; an original, well-integrated design.	Most choices justified; minor errors.	Some design work, but parts are only bought or copied.	Little own design; choices not justified.

D. IMPLEMENTATION (GA5)

D The system is built neatly and safely with suitable tools (CAD, programming, test equipment); good code and wiring practice. (15)

Excellent (12 to 15)	Good (9 to 11)	Adequate (8)	Poor (0 to 7)
Complete, robust and neat; well-structured code under version control.	Complete with small problems.	Works partly; untidy build or code.	Incomplete or unsafe.

E. TESTING, RESULTS AND ANALYSIS (GA4)

E Tests planned from the specifications; measurements recorded with suitable equipment; results analysed, compared with the specifications and discussed honestly. (15)

Excellent (12 to 15)	Good (9 to 11)	Adequate (8)	Poor (0 to 7)
Every specification tested with repeatable measurements and a sound analysis of errors.	Most specifications tested; reasonable analysis.	Some tests; little analysis.	Claims without measurements.

F. TECHNICAL REPORT (GA6)

F Logical structure, clear writing, correct figures, tables, drawings and references; follows the template and page limit. (10)

Excellent (8 to 10)	Good (6 to 7)	Adequate (5)	Poor (0 to 4)
Professional, concise and easy to follow.	Clear, with minor errors.	Understandable but poorly organised.	Hard to follow; many errors.

G. ORAL EXAMINATION AND DEMONSTRATION (GA6)

G A clear presentation; a working demonstration; confident, correct answers that show the work is the student's own. (10)

Excellent (8 to 10)	Good (6 to 7)	Adequate (5)	Poor (0 to 4)
A full working demo; excellent understanding of every part.	A working demo; good answers.	A partial demo; some weak answers.	No working demo or cannot explain the work.

H. PROJECT MANAGEMENT AND PROFESSIONALISM (GA10, GA11)

H Plan followed or sensibly updated; deadlines met; regular meetings with the supervisor; lab book kept; ethical conduct and honest referencing. (5)

Excellent (5)	Good (4)	Adequate (3)	Poor (0 to 2)
Well managed throughout; excellent lab book.	Mostly on time; good records.	Some missed deadlines.	Poorly managed; no records.

TOTAL: 100

END OF MARKING GUIDELINE